

A first Introduction on Toric Varieties

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Abstract. The study of toric varieties is a fascinating area of algebraic geometry because it connects geometry, algebra, and combinatorics in a very visual and concrete way. In this course, we will explore the theory of toric varieties starting from the basics, so only a modest mathematical background is required. The idea is to gradually develop the main concepts and techniques of the subject while understanding why toric varieties have become such an important tool in modern algebraic geometry.

Throughout the course, we will study objects such as cones, lattices, polytopes, and fans, and we will see how they provide a combinatorial description of geometric spaces. One of the most appealing aspects of toric geometry is that many sophisticated ideas can be understood through explicit examples and visual constructions.

As the course progresses, we will move toward more advanced topics and applications of the theory. In particular, we will eventually reach the resolution of certain singularities using Hirzebruch–Jung negative continued fractions, illustrating how toric geometry offers elegant and concrete methods for understanding complicated geometric phenomena. By the end of the course, participants will have a solid introduction to the subject and a view toward more advanced research topics.

Keywords. Toric Varieties, Algebraic Geometry, Convex Geometry.

This course is an introduction to toric geometry with an emphasis on explicit constructions, combinatorial methods, and the study of singularities from the perspective of algebraic geometry. The course will develop the theory progressively, beginning with the combinatorial and algebraic foundations of toric varieties and culminating in the resolution of cyclic quotient surface singularities through Hirzebruch–Jung continued fractions. The general philosophy throughout the course is that many geometric properties of a variety can be encoded by combinatorial objects such as cones, fans, and polytopes.

We begin with the basic combinatorial ingredients underlying toric geometry. The first topics include lattices, dual lattices, semigroups, rational polyhedral cones, dual cones, and convex lattice polytopes. Particular attention will be given to the semigroup associated to a cone and to Gordan’s Lemma, which guarantees finite generation of the corresponding monoid. From these constructions we will define semigroup algebras and explain how affine toric varieties arise naturally as spectra of these algebras. Explicit examples will be studied in detail in order to understand how relations between generators of the semigroup translate into binomial ideals defining affine toric varieties.

The course will then introduce algebraic tori and torus actions. We will study the character lattice and the interpretation of monomials as characters of the torus. Using this perspective, affine toric varieties will be presented as varieties containing an algebraic torus as a dense open subset whose torus action extends to the whole variety. We will discuss the embedding of the torus into affine toric varieties and analyze how the geometry of the defining cone controls properties such as dimension, normality, and singularities.

After the affine case, we will develop the theory of toric varieties associated to fans. A fan will be introduced as a compatible collection of strongly convex rational polyhedral cones, and we will

explain how affine toric varieties corresponding to adjacent cones can be glued together to produce global toric varieties. Fundamental examples such as projective space, Hirzebruch surfaces, and blow-ups of toric surfaces will be discussed from the fan perspective. We will also study the relationship between geometric properties of a toric variety and combinatorial properties of its fan, including completeness and compactness criteria.

An important part of the course will be devoted to the orbit–cone correspondence. We will study the bijection between cones of a fan and torus orbits of the associated toric variety, together with the correspondence between dimensions of cones and dimensions of orbits. This provides a direct combinatorial description of invariant subvarieties and divisors. In particular, we will introduce torus-invariant Weil divisors, canonical divisors, and the divisor group generated by the rays of the fan.

The next section of the course focuses on singularities and resolutions in dimension two. We will define smooth cones and smooth fans and prove the criterion stating that a toric variety is smooth if and only if all cones in the corresponding fan are smooth. Since singularities of toric surfaces occur precisely at torus-fixed points corresponding to two-dimensional cones, this reduces the problem of resolution to a local combinatorial procedure.

We will then study toric morphisms and subdivisions of fans, leading to the construction of resolutions of singularities. In dimension two, we will discuss the explicit algorithm for resolving a cone by subdividing it into smooth cones using lattice points lying on the boundary of the convex hull associated to the cone. This construction yields the minimal toric resolution of a surface singularity. We will analyze the exceptional divisors appearing in the resolution and compute their self-intersection numbers directly from the combinatorics of the subdivision. Special attention will be given to configurations of exceptional curves and their interpretation as chains of smooth rational curves.

Once resolutions are understood from the fan perspective, we will study cyclic quotient singularities. Every two-dimensional toric singularity can be expressed as a quotient singularity of type

$$\frac{1}{d}(1, k),$$

and we will explain how the cone determines the quotient structure through lattice indices and normal forms. The course will discuss the relation between the determinant of the cone generators and the order of the finite cyclic group acting on the affine plane. Several examples, including singularities of type A_n , will be developed explicitly.

The final part of the course will focus on Hirzebruch–Jung continued fractions and their role in the minimal resolution of cyclic quotient singularities. We will explain how a singularity of type

$$\frac{1}{d}(1, k)$$

admits a minimal resolution whose exceptional locus is determined by the negative continued fraction expansion of

$$\frac{d}{k}.$$

The coefficients of this expansion determine the self-intersection numbers of the exceptional curves appearing in the resolution. Explicit computations and examples will be carried out in detail to illustrate how the combinatorics of continued fractions corresponds geometrically to subdivisions of cones and configurations of exceptional divisors.

The course will also cover several aspects of toric birational geometry and Fano varieties. We will study simplicial fans, quotient singularities, canonical and terminal singularities, and the combinatorial characterization of toric Fano varieties through Fano polytopes. In dimension two, this naturally leads

to the study of toric del Pezzo surfaces. The dictionary between geometric properties and combinatorial conditions on the associated polytope will be emphasized throughout this part of the course.

Another important topic will be the relation between divisors and polytopes. Given a torus-invariant divisor, we will associate a polytope in the dual lattice and study how geometric properties of the divisor are reflected in the combinatorics of the polytope. We will discuss global sections, ampleness, lattice volume, and normal fans of polytopes, together with examples involving projective spaces and Hirzebruch surfaces. This point of view provides a geometric interpretation of divisors and embeddings of toric varieties into projective space.

Finally, depending on the pace of the course, we may discuss weighted projective spaces, weighted blow-ups, and the quotient description of toric varieties through Geometric Invariant Theory. Topics such as weight matrices, secondary fans, irrelevant ideals, and stability conditions will be introduced in examples in order to illustrate how toric varieties can also be described as quotients of affine spaces by torus actions.

The main references for the course will be [2] by William Fulton and [1] by David Cox, John Little, and Henry Schenck, together with supplementary notes and articles related to singularities and resolutions in the toric setting.

References

- [1] *D. Cox, J. Little, and H. Schenck*, Toric varieties. Providence, RI: American Mathematical Society (AMS) (2011; Zbl 1223.14001)
- [2] *W. Fulton*, Introduction to toric varieties. The 1989 William H. Roever lectures in geometry. Princeton, NJ: Princeton University Press (1993; Zbl 0813.14039)